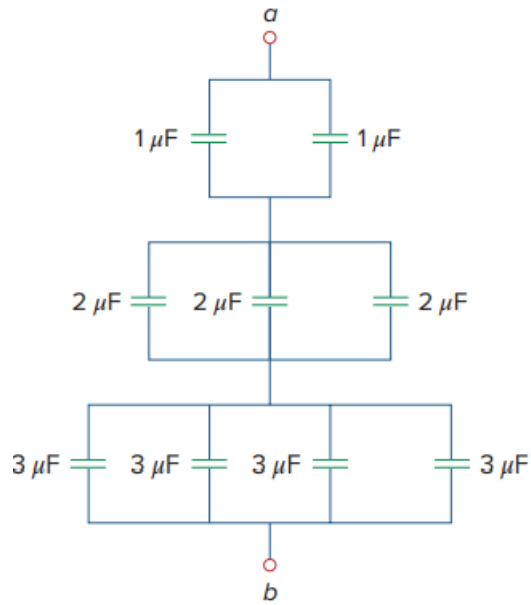


Problem

Find the equivalent capacitance at terminals a - b of the circuit in Fig. 6.54.

Fig. 6.54:



Step-by-step solution

Step 1 of 2

Consider the following circuit consisting of capacitances connected as shown in Figure 1.

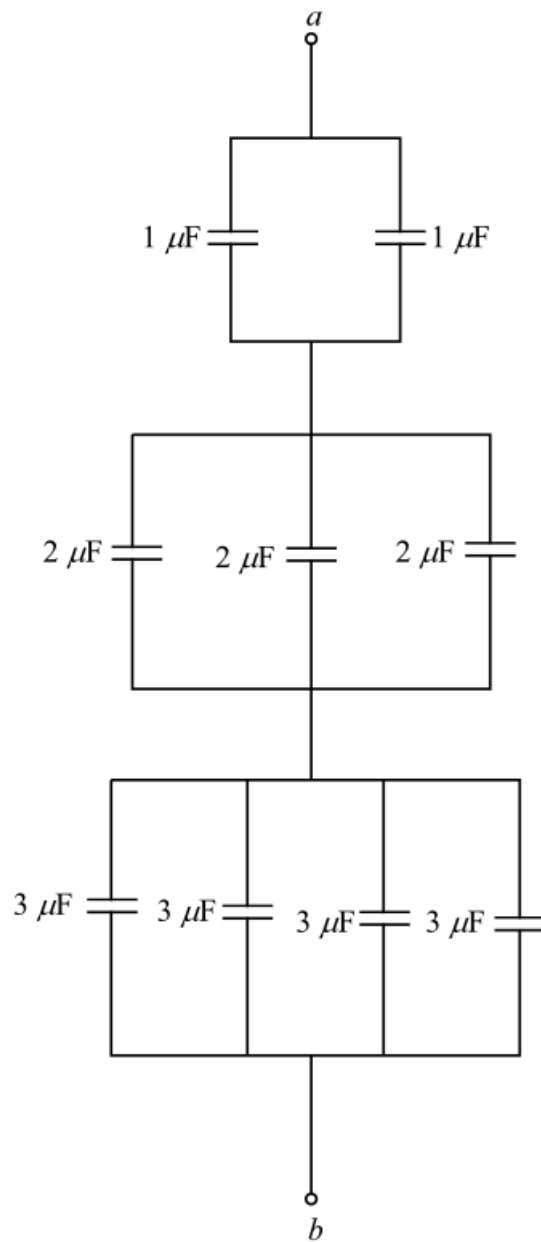


Figure 1

Determine the equivalent capacitance of the parallel combination of capacitances.

$$\begin{aligned}C_{eq1} &= 1 \mu\text{F} + 1 \mu\text{F} \\ &= 2 \mu\text{F}\end{aligned}$$

$$\begin{aligned}C_{eq2} &= 2 \mu\text{F} + 2 \mu\text{F} + 2 \mu\text{F} \\ &= 6 \mu\text{F}\end{aligned}$$

$$\begin{aligned}C_{eq3} &= 3 \mu\text{F} + 3 \mu\text{F} + 3 \mu\text{F} + 3 \mu\text{F} \\ &= 12 \mu\text{F}\end{aligned}$$

These three capacitances are in series. Determine the equivalent series capacitance.

$$\begin{aligned}\frac{1}{C_{eq}} &= \frac{1}{C_{eq1}} + \frac{1}{C_{eq2}} + \frac{1}{C_{eq3}} \\ &= \frac{1}{2 \mu\text{F}} + \frac{1}{6 \mu\text{F}} + \frac{1}{12 \mu\text{F}} \\ &= \frac{9}{12 \mu\text{F}} \\ &= \frac{3}{4 \mu\text{F}}\end{aligned}$$

$$\begin{aligned}C_{eq} &= \frac{4}{3} \mu\text{F} \\ &= 1.33 \mu\text{F}\end{aligned}$$

Thus, the equivalent capacitance at terminals *a-b* of the circuit is 1.33 μF .